

The Effect of Rejection Friction on Repeated Cookie Consent Decisions

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Abstract—Cookie consent banners repeatedly appear across websites, requiring users to make frequent privacy-related decisions. Prior work has shown that interaction costs can influence cookie consent choices, but less is known about how such effects change across repeated consent decisions. This study investigates how rejection friction affects users’ privacy choices when cookie consent requests are repeatedly presented. In a between-subjects experiment, 20 participants were randomly assigned to either an Easy Reject condition, in which a “Reject All” button was immediately available, or a Hard Reject condition, in which rejection required navigating through cookie settings. Participants made cookie-related decisions on 12 fictitious websites. Although rejection rates were similar between the two conditions during the initial trials, the rate of “Reject All” selections decreased in the Hard Reject condition as the experiment progressed. A significant interaction was observed between condition and exposure order. These findings suggest that rejection friction may exert a stronger influence over repeated consent decisions and highlight the importance of making privacy-protective options readily accessible in cookie consent interfaces.

Index Terms—cookie consent, privacy, rejection friction, repeated decisions, user interface

I. INTRODUCTION

Modern Internet users visit numerous websites for information seeking, shopping, finance, and communication, and frequently encounter cookie consent banners in the process. Cookies support useful web functions such as maintaining login states and user preferences, but they are also used for behavioral advertising and user tracking, making them an important privacy concern [1]. In response, the European Union’s ePrivacy Directive generally requires user consent when information is stored on, or accessed from, a user’s terminal equipment [2]. The General Data Protection Regulation (GDPR) further defines consent as a freely given, specific, informed, and unambiguous indication of the data subject’s wishes [3].

In practice, however, users do not make a cookie consent decision only once. Across different websites, they may repeatedly face similar privacy choices such as “Accept All,” “Reject All,” and “Manage Cookies.” Utz et al. found that cookie consent notices appeared on more than 60% of popular European websites in their post-GDPR field study and discussed the burden associated with widespread privacy notices [1]. Under repeated consent requests, users may increasingly process banners quickly rather than carefully reviewing their contents, or may perceive additional privacy-protective interactions as

burdensome. In particular, when rejecting cookies requires entering a settings interface and completing additional steps rather than a single click, it is important to examine whether the effect of this interaction cost changes over repeated decisions.

Prior security and privacy HCI research has shown that cookie choices can be influenced by interface factors including choice architecture, visual emphasis, wording, and the accessibility of rejection options [1], [4]–[6]. Nouwens et al. reported that removing an opt-out button from the first page increased consent by 22–23 percentage points [4]. Congiu et al. showed that the visual prominence and wording of rejection controls can affect rejection behavior; in a condition that offered a customization option instead of a direct rejection button, only 3% of participants rejected non-essential cookies [5]. Papenmeier et al. further found that the effort required to make a cookie choice and button highlighting influenced users’ decisions across repeated interactions with 12 cookie banners [6].

Building on this work, we combine rejection friction with repeated cookie consent requests and examine whether the effect of rejection friction changes as repeated exposure progresses. Participants sequentially visited fictitious websites under either an Easy Reject condition, where “Reject All” was directly available on the first screen, or a Hard Reject condition, where rejection required navigating through a cookie settings screen. We recorded the final cookie choice, rejection rate, decision time, and number of interactions for each trial, and separately measured subjective fatigue after the experiment. Our primary analysis focuses on the interaction between rejection friction and exposure order to characterize behavioral change over repeated consent decisions.

II. BACKGROUND

A. Cookie Banner Design and Rejection Friction

Users’ cookie choices can be shaped not only by their privacy preferences but also by how choices are presented. Nouwens et al. showed that removing an opt-out button from the first page increased consent by 22–23 percentage points, while providing more granular choices decreased consent by 8–20 percentage points [4]. Bauer et al. distinguished between *choice-making architecture*, which encourages users to make an explicit choice, and *choice-outcome architecture*, which steers users toward a particular outcome; their experiments

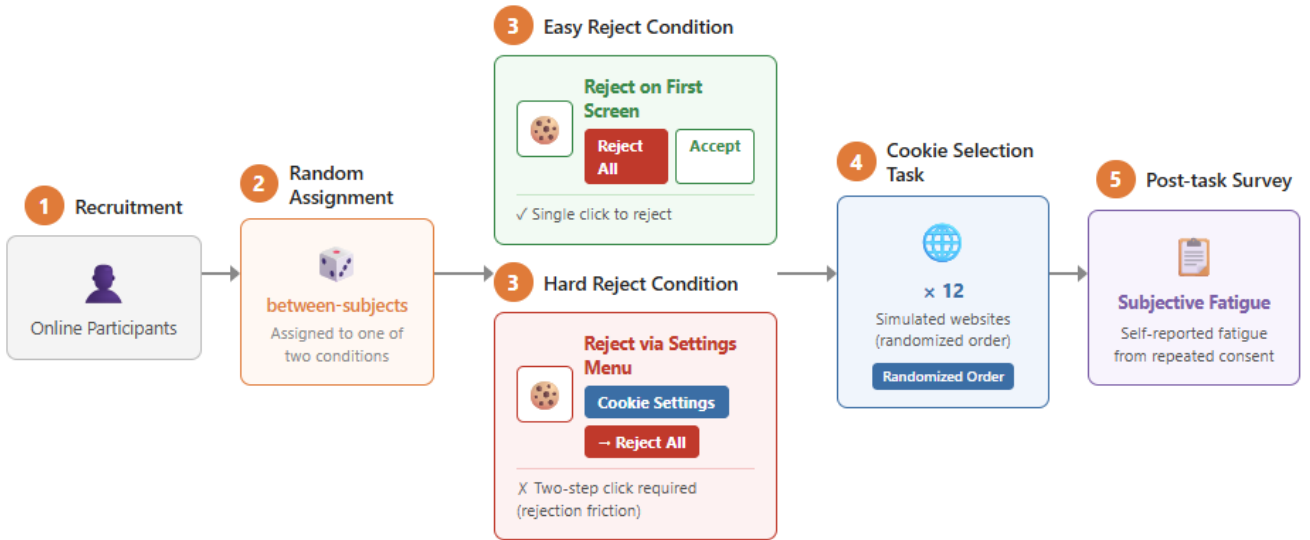


Fig. 1: Experimental design and procedure. Participants were recruited online and randomly assigned to either the Easy Reject or Hard Reject condition. Participants then completed cookie selection tasks on 12 simulated websites presented in a randomized order. Finally, they answered a post-task survey measuring subjective fatigue.

showed that such design manipulations could increase consent rates by up to 17 percentage points [7].

Congiu et al. also demonstrated that visual emphasis and wording of rejection controls can affect rejection behavior [5], while Papenmeier et al. found that interaction effort and button highlighting influence cookie choices [6]. Based on these findings, we use the term *rejection friction* to denote additional interaction required to reach a privacy-protective rejection choice. We operationalize this concept by comparing an Easy Reject condition, in which “Reject All” is directly available, with a Hard Reject condition, in which participants must first enter “Cookie Settings” before selecting “Reject All.”

B. Repeated Cookie Consent Decisions

Cookie decisions are repeatedly encountered across websites during ordinary web use. Papenmeier et al. asked participants to make choices across 12 cookie banners and found that effort and highlighting affected decisions, while approximately two-thirds of participants exhibited consistent acceptance or rejection behavior [6]. This suggests that interface factors and individual choice tendencies can jointly shape behavior even in repeated settings.

In this study, repeated exposure is not treated as equivalent to fatigue. Instead, *Exposure Order* represents behavioral change over the sequence of trials, whereas subjective fatigue is measured separately using a post-experiment questionnaire.

C. Research Gap

Existing studies establish that cookie banner design and interaction effort can influence cookie choices [4]–[7]. We focus specifically on whether the effect of rejection friction changes over a sequence of repeated consent decisions. We

therefore analyze the interaction between Rejection Friction and Exposure Order to determine whether the behavioral difference between Easy Reject and Hard Reject changes as repeated exposure progresses.

III. METHOD

A. Experimental Design and Procedure

We conducted a between-subjects experiment in which participants were assigned to either the Easy Reject or Hard Reject condition (Fig. 1). Each participant repeatedly made cookie choices on 12 fictitious websites presented in randomized order and then completed a post-experiment questionnaire. In the Easy Reject condition, participants could select “Reject All” with one click from the initial banner. In the Hard Reject condition, participants had to navigate through “Cookie Settings” before selecting “Reject All,” thereby introducing rejection friction (Fig. 2).

B. Measures and Analysis

The primary dependent variable was whether “Reject All” was selected in each trial. We additionally recorded the final choice, decision time, number of clicks, and whether the participant entered the settings screen. The presentation position of each banner was defined as *Exposure Order*. We examined the Rejection Friction $\times$ Exposure Order interaction to test whether the difference in rejection behavior between conditions changed over repeated trials. Subjective fatigue was measured separately in the post-experiment questionnaire.

IV. RESULTS

We analyzed 240 trials from 20 participants: 11 in the Easy Reject condition and 9 in the Hard Reject condition. The overall “Reject All” rate was 56.8% in the Easy Reject

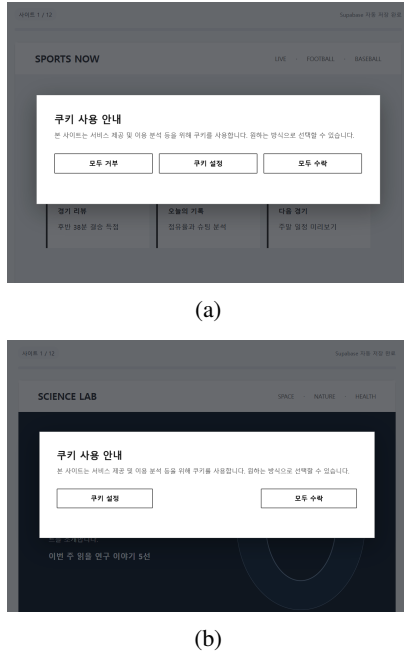


Fig. 2: Cookie consent banners used in the experiment. (a) Easy Reject provides a direct “Reject All” option. (b) Hard Reject requires navigating through “Cookie Settings” to reject cookies.

condition and 40.7% in the Hard Reject condition. During the initial trials (1–3), rejection rates were similar between conditions (Easy: 51.5%; Hard: 51.9%). During the final trials (10–12), however, the Easy Reject condition remained at 54.5%, whereas the Hard Reject condition decreased to 18.5% (Fig. 3). A repeated-measures logistic generalized estimating equation (GEE) analysis showed a significant Condition $\times$ Exposure Order interaction ($\beta = -0.202$, OR = 0.817, 95% CI [0.709, 0.942], $p = .005$). This interaction indicates that the difference in rejection behavior between the two conditions changed over the sequence of repeated trials. The mean number of clicks was higher in the Hard Reject condition (2.15) than in the Easy Reject condition (1.15). Median decision time was also higher in the Hard Reject condition (1.43 s) than in the Easy Reject condition (0.78 s). In contrast, subjective fatigue was high in both conditions (Easy: 6.64/7; Hard: 6.58/7), with no significant difference between conditions ($p = .811$).

V. DISCUSSION

The Easy Reject and Hard Reject conditions showed nearly identical “Reject All” rates during the initial trials, but rejection in the Hard Reject condition decreased as repeated exposure progressed. This pattern suggests that the influence of an interface requiring additional interactions for rejection may be limited in a single decision yet become more pronounced across repeated cookie consent decisions. Evaluations of cookie consent interfaces should therefore consider not only whether a rejection option exists on an individual banner, but

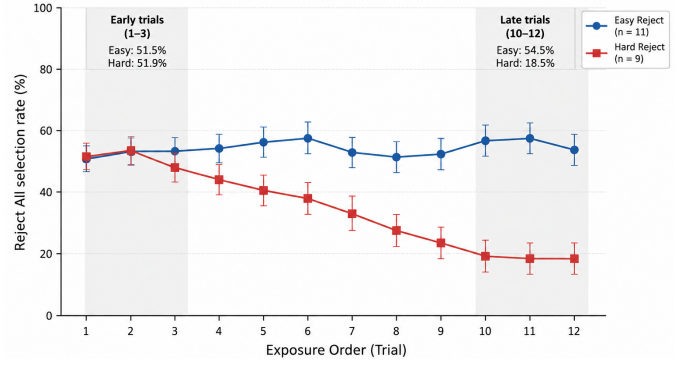


Fig. 3: Reject All selection rates across exposure order for the Easy Reject and Hard Reject conditions. Shaded regions indicate the early (Trials 1–3) and late (Trials 10–12) periods.

also how users can access that option when similar decisions recur across multiple websites. The higher number of clicks and longer decision time in the Hard Reject condition further indicate that the additional settings step imposed measurable interaction costs. However, subjective fatigue was high in both conditions and did not significantly differ between them. The observed decline in rejection under the Hard Reject condition therefore cannot, based on the present results, be directly attributed to greater self-reported fatigue. Further work is needed to disentangle how repeated interaction costs, accessibility of rejection options, and fatigue jointly influence privacy choices over time.

Limitations. This study has several limitations. First, the sample was limited to 20 participants, and the results should therefore be generalized to broader web-user populations with caution. Second, the experiment used fictitious websites in a controlled setting, so additional validation is required to determine whether the same behavior occurs during real-world browsing. Third, repeated exposure consisted of 12 cookie decisions; we did not test whether the same pattern persists over longer or more varied sequences. Finally, although both conditions produced high subjective fatigue ratings, there was no significant between-condition difference. The decrease in rejection observed in the Hard Reject condition therefore cannot be directly explained as an effect of increased subjective fatigue.

VI. CONCLUSION

This study examined how the effect of rejection friction on privacy choices changes under repeated cookie consent requests. Rejection rates were similar between the Easy Reject and Hard Reject conditions at the beginning of the experiment, but “Reject All” selections decreased in the Hard Reject condition as repeated exposure progressed. These findings indicate that additional interaction costs associated with cookie rejection may continue to shape privacy choices across repeated consent situations. Cookie consent interfaces should therefore provide privacy-protective options that remain readily and

consistently accessible, rather than merely making rejection technically available.

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